Chapter 2

A PC-based physiological measurement and data acquisition system

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Abstract

The portable physiological measurement system that will be discussed here enables the user to perform EEG and other physiological measurements even outside the laboratory environment. To maintain a complete, portable measuring disposition at a low cost, it was decided to realize the acquisition and data storage by means of a standard PC. Compact extendable amplifiers have been realized by reducing the number of adjustment options. Remote control of the measurement amplifiers and real-time data preprocessing enable the user to carry out these adjustments using a PC. These preprocessing operations are executed via a digital-signal processor (DSP) mounted on a special data acquisition board, thus liberating the PC itself from these operations. The main tasks and operations of this board are sampling, event collection, data reduction (preceded by effective digital filtering), and the computation of differences between channels in order to make the EEG derivations. Furthermore, the DSP is able to execute rapidly 'off-line' signal processing such as FFTs, averaging, etc. on data that has been already collected. The PC itself has a data storage and a signal monitoring function. The acquisition board is controlled by a powerful series of commands, which can be down-loaded and that can be automatically generated by menu driven software enabling the user to choose or create the application needed without any difficulty.

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1. Introduction

EEG measurements used for psychological research are mostly carried out either in a laboratory environment or in a hospital environment. There are a number of reasons why such measurements are not usually executed at the actual place of work in a factory or at some other laboratory-independent location. The first reason deals with the nature of the research and the applicability of evoked-response (ERP) research in a real working environment; until recently theoretical applications of this technique were not possible for complex practical situations (like in the working place, schools, nursing homes, traffic, etc.). However, there has been a rapid change; there is an increasing need for EEG measurements carried out outside the laboratory environment. The other two reasons deal with technical aspects: the size of the apparatus (polygraph, computer for data acquisition, tape units for data storage), and the fact that the measurements are sensitive to external disturbances (requiring the electrical shielding of the measurement room) make it almost impossible to carry out measurements outside the laboratory environment.

Recent technical advances such as the development of small noise-insensitive physiological measurement amplifiers and the development of data acquisition facilities on a PC with large data storage capacity make it possible to develop a transportable measurement system that enables the user to take measurements in various situations outside a shielded room.

This measurement system, composed of both an amplifier part and an acquisition part, will be discussed. Further, a measurement disposition to which the present measurement system has been applied will be described.

2. The amplifier system

The amplifier system consists of a basic unit and one or more optional extensions units. The basic unit includes eight EEG channels (Figure 1a) in addition to one 1 EOG channel, two EMG channels, and one ECG-channel for more general applications (Figure 1b).

To this basic system, one or more extension units of 16 EEG channels each can be connected. These extension units can also be used separately or connected together. When connected to the basic unit or to another extension unit, the control and operation are directed by the preconnected unit. For the EEG channels, a 'common average' and a 'common reference' derivation system has been created. EEG channels from connected units can be switched off from the basic unit by choosing their own derivation procedure and sensitivity adjustments. In this way the basic unit can, for instance, be used to create additional EOG channels.

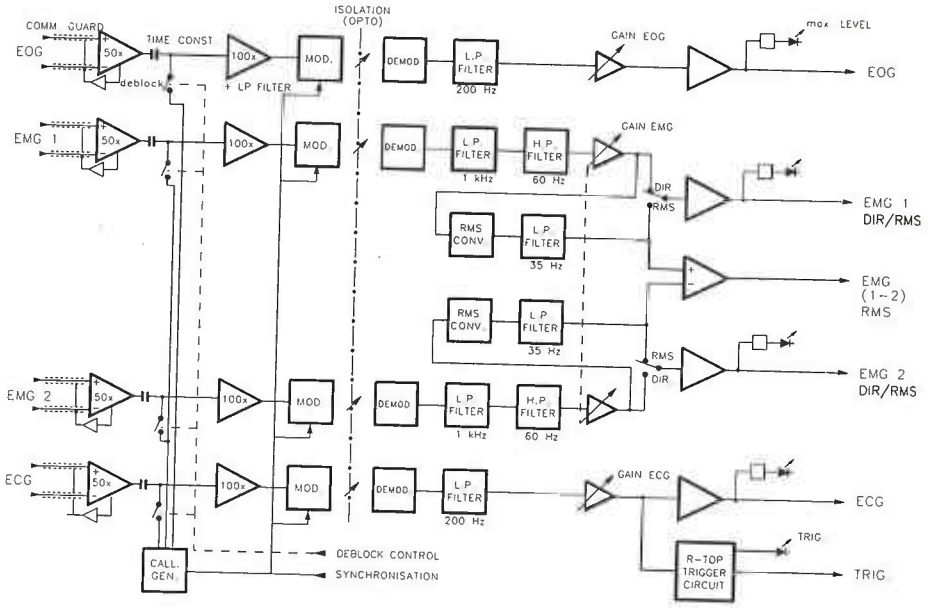


Figure 1a. Simplified block diagram of the EEG channels.

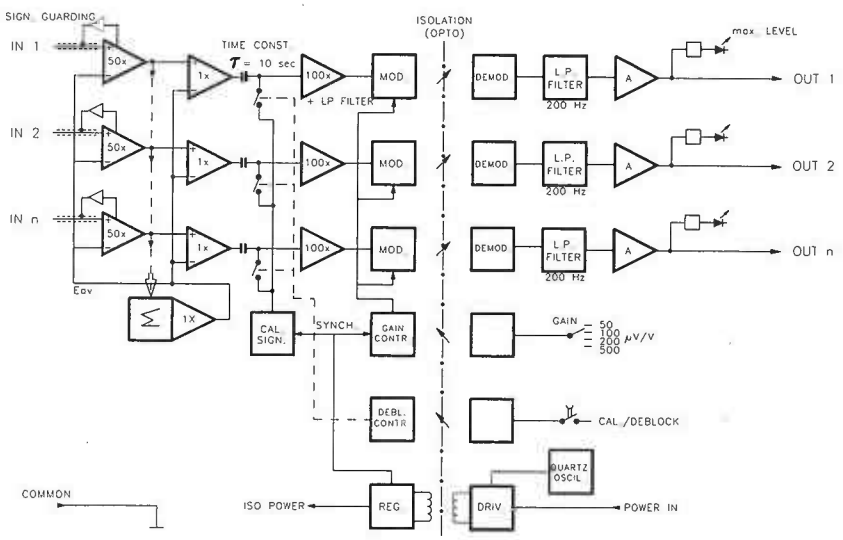


Figure 1b. Simplified block diagram of the EOG, EMG, and ECG channels.

2.1. Working principle and possibilities of the channels

- *EEG*. The main principle of the amplifier system is that the average voltage of all the connected preamplifiers is fed back to the negative input of each preamplifier. This principle only amplifies the electrode input voltage; the common signal is not amplified (Vos, Lammertsma, & van Eykern, 1977). Because of the availability of this average value, it is possible to subtract the common signal from the outputs of the preamplifiers. The electrode derivation E_i - E_{av}, thus obtained, is called the 'common average' (CA), in which E_{av} is the average of all inputs and Ei the input voltage of input i.
  After each preamplifier, there is a time constant to dispose the signals of offset voltages (mainly the already amplified electrode-potential differences). After this step, the signals are further amplified and modulated (pulse width modulation). The modulated - and now digital - signals are transmitted optically to the demodulators (galvanic separation).
  The low-pass filters which generally comprise the demodulators, also serve as anti-alias filters for the analog/digital conversion in the PC. At this point, the signals are still in the common-average mode (see above). This is one of the electrode derivation methods that can be selected. In this method, the electrode potentials are derived against the average of the potentials of all actively connected electrodes. A second possibility is that channels can be reflected in the so-called common-reference mode (CR). The first channel of the system (in the CA mode) will then be subtracted from the other channels by use of differential amplifiers. Often one of the ear lobes or the mastoid is used as a reference during EEG registrations. If desired, it is also possible to use both ears as a reference. In this case, the basic unit offers the possibility of determining the electric average of the first two channels, which then serves as a reference.
  If all channels are reflected in the same derivation mode, any derivation of the mutual channels can be made, either by external differential amplifiers or by 'on-line' or 'off-line' calculations in the computer.
- *EOG*. There is a standard differential EOG channel independent of the EEG channels.
- *EMG*. The two differential EMG inputs are independently adjustable for direct high band-width recording, or recording in RMS envelope mode (Root Mean Square, i.e. instantaneous power of the signal).
- *ECG*. A precision R-peak trigger has been connected to the ECG-channel in order to register R-peak interval times from the ECG. This trigger automatically adjusts to the ECG amplitude. The trigger output can be recorded as a digital event.

The fixed-time constants in the amplifier system are jointly deblocked. During the time that the deblocking button is held down, a calibration signal is generated at all channels, viz. a 1 Hz block signal of 100 µV top-top for EEG

and EOG; 1 mV top-top for ECG; for EMG a 100 Hz sinus signal of 100 µV rms is used. After release of this button, the time constants are deblocked for 1 second. Both functions of this button can also be activated using the remote mode, controlled by the PC during the measurements.

All channels have an out of range indication of +/- 1 Volt for the registration equipment. In case this limit is exceeded, a LED will light up at the relevant channel and a digital pulse is given as an alarm.

2.2. Noise insensibility

Measuring voltages in the order of µVolts requires amplifiers with specific qualities, especially if the amplifiers are used under awkward conditions, as is the case with EEG measurements made outside of a laboratory setting.
The following disturbances could be encountered.

- Strongly varying signal sources and fluctuating impedances of the electrodes.
- Signals containing communal noise.
- Electric and magnetic noise fields.

Further the following special techniques and high quality specifications should be required to avoid artefacts in the measurements.

- High impedance inputs 10^{12} Ohm.
- Low noise specifications 2 µV top-top.
- Large CMR (Common Mode Rejection) 110 dB at 50 Hz.
- Large IMR (Isolation Mode Rejection) 170 dB at 50 Hz.
- Guarding of the electrode leads (active shielding).
- Floating measuring method by means of an adequate galvanic separation; leakage current at 50 Hz lower than 5 µA, inducing a low common voltage.
- Compact and symmetric construction (simplifying internal shielding and partly making them superfluous).

'Guarding', also called 'bootstrapping' is an active way of shielding during which the same signal is connected both at the shielding (via a driver) and at the input lead. This method of shielding yields a good CMR of AC signals and does not contribute to producing cable-movement artefacts.

Another important point in this framework is the synchronization of all internal oscillators; this avoids interferences so that measurement signals are not disturbed. From crystal oscillator, frequencies are derived for the HF power transfer, the modulation frequency, and the calibration signals. Much care has been bestowed on the above mentioned issues in order to get a high quality amplifier with regard to the technical aspects of measuring, thus enabling EEG measurements to be made under various conditions.

2.3. Safety

Much attention has also been given to safety. Using an optical signal transfer,

directly following pre-amplification, results in complete electrical separation between the subject and the outputs and power supply of the measurement unit. Any other electric apparatus can be connected to these outputs. Power supply for the amplifiers in the preamplifier stage occurs via HF power transfer. The separation isolation withstands high voltages and has a very low leakage current at 50 Hz. Currents caused by any possible defect in the preamplifiers are limited since at each electrode input resistances have been included. Without electric separation, a measurement disposition is only safe when the measurement amplifiers themselves and all connected as well as surrounding apparatus of the electric installation in the measurement room are safe, both separately and coherently. This means expensive measures and periodical control. Any addition of apparatus to the measurement disposition may influence the electric safety.

The design chosen here makes these extra safety measures unnecessary and offers more opportunities to perform measurements in any room.

2.4. Portability

The measurement amplifiers produced were required to be compact (18.5x11x7 cm), and therefore, a some what stripped set-up was chosen. The number of adjustment options has been kept limited by making such fixed choices, as time constants, low-pass filters, and a selection of EEG derivations. Within a measurement group, the user is not able to select different sensitivities per channel. A great advantage of this limitation, however, is that there are only minor differences in signal transfer (matching of the filters, equal amplification, etc.). The operation of such a compact system has been kept plain, and errors with regard to selected adjustments do not occur easily. Also in view of production costs, this framework is advantageous. These apparent limitations in operating possibilities may for a large part compensated for by 'real-time' options for signal processing in the PC 's data acquisition system.

Regarding band widths, the measurement amplifiers have a large frequency range and are supplied with fixed- frequency, low-pass filters. This framework has been decided on to enable the user to sample with a relatively high and fixed frequency (adapted to the filters). It is then possible to reduce sampling-frequency by software after suitable digital filtering.

3. The acquisition system

On the basis of an intelligent-interface board (available from Microstar* Laboratories, inc.), a data acquisition system has been constructed that can be implemented in a PC-AT. This acquisition board has been equipped with a 16-bits processor plus a DSP (Digital Signal Processor) and a real-time multi-

tasking operating system. Besides signal collecting (A/D conversion) the acquisition board enables the user to perform such signal processing as digital filtering, fast fourier transforms, averaging, and peak detection. The applications can be developed without any programming of the board itself; compiling a series of commands is the only necessity. The acquisition processor receives commands from the PC-application program or directly from the keyboard. The board itself has a (limited) data-buffering capacity. Processes, once started, run autonomously without imposing any load on the PC; as a result, the PC is available for user interactions, data storage, and signal monitoring.

The following I/O specifications can be given.

Inputs:	analog:	16 channels (expansion until 512)
	digital:	16 bits (expansion until 128).
Outputs:	analog:	2 channels (2 separate D/A conversions)
	digital:	16 bits.
A/D conversion:		12 bits resolution, 156 ksamp./sec.; amplification to be programmed and to be selected per channel.

3.1. Signal sampling

When sampling continuous-analog signals that are to be converted into discrete digital-time series, attention should be paid to the following important issues: simultaneous and equidistant sampling and anti-aliasing.

Simultaneous sampling
This is important in order to prevent phase errors between the channels. These errors occur when applying the (most frequently used) sequential-sampling method. Simultaneous sampling is obtained by applying a so-called 'sample and hold' in each channel. A good alternative for such an expensive and space consuming solution would be to rapidly sample the channels sequentially and to repeat this process at fixed interval times determining the real sample frequency. The time differences between samples of different channels are then relatively so close, that for all practical purposes, one can speak of simultaneous sampling. This option has been chosen for in the described data-acquisition board.

Equidistant sampling
This is guaranteed as the sampling in the acquisition board is an autonomic process which is not affected by unpredictable interrupts in the host computer. In the present set-up, no problems of this type are to be expected.

Anti-aliasing
If signals with a frequency above half the sample frequency are present then again errors may occur. High frequencies are then presented in lower frequency

ranges; this is called *aliasing*. Theoretically, such errors only disappear completely when the least-significant bit of the AD convertor is no longer affected. To avoid this problem, it is recommended to always use so-called anti-alias filters so that the cut-off frequency is permanently connected to the sampling frequency.

For the measurement system, a fixed sampling frequency of 1000 Hz was chosen which offers the advantage of using anti-aliasing filters with a fixed cut-off frequency. For that reason, the measurement amplifiers are supplied with 200 Hz low-pass Butterworth filters, with an attenuation of -40 dB at 500 Hz. Since big noise sources are usually not present in the natural attenuation frequency of the EEG-frequency spectrum, the desired attenuation at 500 Hz can easily be reached. At this sampling rate, the EMG signals have to recorded in RMS mode (band width: DC - 30 Hz). In the direct mode (60 - 1000 Hz band width), the sampling rate has to be at least 3000 Hz.

3.2. Data reduction

The 200 Hz band width of the measurement amplifiers is too large for most research. EEGs are usually measured with a maximum band width of 30 Hz. The 1000 Hz sampling of the measurement system would yield too much data. For example, a reduction of a factor of 10 can be easily reached by selecting the first sample out of 10 consecutive data points from each channel. In this way a sampling frequency of 100 Hz is reached. However, in order to prevent aliasing in this set-up a digital low pass filter has to be applied first.

The here used FIR filter (Finite Impulse Response) offers enormous advantages with respect to its analog counterpart. First, without any additional cost a very great filter steepness is reached; more filtering sections only require more calculation time. Secondly, the FIR-filter shows a perfectly linear phase-characteristic, and thirdly, the cut-off frequency can be altered with great flexibility. It has become evident that the processor on the acquisition board, together with other tasks still to be performed, is able to perform a 101 tap, 30 Hz low-pass FIR-filtering task for all 16 channels in real-time at a 1000 Hz sampling frequency per channel.

In order to record EMG channels in the direct mode with an adequate sampling frequency, the best choice is to sample all channels to be recorded at the same high frequency (e.g. 3000 Hz). By using different FIR-filters, with unequal cut-off frequencies, it is possible to obtain an adequate effective sampling rate for each of the channels.

3.3. Event recording

In order to register digital events, such as an ECG trigger and stimulus synchronization pulses, the acquisition board can be used to perform another

important task: event recording (Mays & McElligott, 1976). The board's 16-bit digital-input channel is sampled synchronously with the analog channels at 1000 Hz. The value of this digital channel is compared to the previous sample. If one or more bits has been altered (indicating the detection of an event), the bit code is sent to the host computer, together with the time of occurrence (clock value in ms). As sampling of both the analog and digital channels are being controlled by the same clock in the data acquisition board, the synchronization of these data flows does not give any difficulty (Mulder, Gerritsma, & van den Hoven, 1985). Synchronization with external data flows (including stimulus presentation) is achieved in this design by connecting a synchronization pulse (stimulus, start, stop, etc. from the stimulus presentation unit) to the data-acquisition system.

3.4. Expansion

The acquisition board supports a maximal expansion of the number of channels using multiplexers of 512. With a maximal sampling frequency of 156 ksamp/s, it is then theoretically possible to sample 156 channels at 1000 Hz. However, with regard to the total processing time needed to perform the data processing, in particular for digital filtering, expansion in the number of channels will result in reduced filter performance; it will be necessary to select a filter with fewer taps, and consequently with a less steep-filter characteristic. However, even up to 48 channels, filter performance is obtained which is better than that of most analog anti-aliasing filters currently applied. Another possible way of achieving expansion is to install several acquisition boards in the PC. The boards will then be coherently synchronized using a connector intended for this purpose. The software driver for this expansion has been prepared.

3.5. Programming of the acquisition board

The PC offers software menus enabling the user to choose or compose applications in a simple way. From the PC, a powerful series of commands is automatically sent to the acquisition board via various software modules for command- and parameter generation. Also, user programs can be written in Pascal or another higher programming language, in which the commands for the acquisition board can be included. These commands control the real-time, multi-tasking firmware from the acquisition board. Several types of commands are available. Most of them are supplied with one of more of the following parameters.

- System commands: start, stop, system options, status, etc.
- Input commands: channel configurations, sampling frequency, triggering.
- Processing commands: averaging, filtering, FFT, variance, etc.

- Output commands: output buffers, data transfer, data format.

With over a 100 commands available, tasks can be developed in which several processes can be run at the same time, either in combination or independent of each other.

Another nice option of the processing board includes the possibility of processing data collected in an earlier stage. Data can be sent from the PC to the board in order to enable fast signal processing (digital filtering, FFT, peak detection); results can be returned to the PC for further analysis and/or signal presentation.

4. The PC system

The host PC system has been developed as a digital data-recorder offering a signal-monitoring function. By giving the application program in the PC a data-transfer function (from acquisition board to disk memory) applying exchangeable (possibly optical) disks and/or network facilities and implementing an additional signal-monitoring function in the PC application program, a flexible and reliable data collection station is created that will make analog back-ups redundant in most applications.

Because of the many possibilities of the data acquisition board, the PC itself has almost nothing to do with either sampling the analog and digital channels or with data processing. This makes it possible to monitor 16 channels continuously in real time. It is possible to change the number of displayed channels and the selection of channels while executing the program. Additionally, scaling for the x- and y-axis can be adapted. Another available feature is the possibility of displaying recorded events by representing these by colored arrows on the screen, on the same time-axis as the sampled analog-channels. The same options can be used off-line with previously recorded data files. The 'Print Screen' command can be used to obtain a hard copy on paper.

In order to obtain an adequate file format (TEMPUS-format, Mulder, Gerritsma, & Van der Hoven, 1985), the data are sorted in blocks of 128 samples before writing to disk. All necessary information concerning sampling rate, channel numbers, start time of sampling, calibration etc. are stored in a file header. Digital events are stored in a separate ASCII file, as well the recording channel number and event occurrence time. Synchronization information is stored in the header of the sampled data-file.

One last important point to be mentioned is that in order to prevent data loss, recording data to disk always has a higher priority than signal monitoring.

5. Example of a measurement disposition

The electrophysiological measurement system discussed here, is composed of the following components.

- EEG measurement amplifiers, with channels for EOG, EMG, and ECG.
- Data acquisition board including data-reduction and data-processing possibilities.
- PC system enabling data recording and signal monitoring.

The measurement system fits very well in an ERP-research environment which uses the PC to present the stimulus, but of course, it can also be used in other situations (Figure 2).

Because of the electrical separation, the measurement amplifiers can be connected directly and safely to the acquisition system. The measurement amplifiers have been installed near the subject; the amplifier channels are automatically switched off if the electrode input has not been plugged in. The measurement amplifiers are powered from the main by a simple power unit. The acquisition board in the PC samples the channels and takes care of the data reduction. The amount of reduction depends on the desired frequency contents of the data. The amplifier system can be remote controlled by using the digital outputs of the acquisition board. Digital signals, such as those of response panels and measurement amplifiers (e.g., the ECG trigger and the signal for the out-of-range alarm) are registered as events (1 ms resolution). Synchronization of the acquisition process with the stimulus presentation, also occurs through this event registration (via one of the acquisition board's digital inputs). The collected and preprocessed data are put on a diskette by the PC, or via a data-communication line (e.g., Ethernet) are sent to a mainframe. During this storage process, signal monitoring is possible, to ensure adequate registration. The acquisition process and stimulus process are controlled via the PC keyboard. Expansion of the number of EEG channels is possible by connecting more amplifier units. Several acquisition boards can be installed and coherently synchronized to the PC. In addition, signals from other measurement apparatus such as blood pressure or respiration can be simultaneously recorded.

Other useful applications

Electrophysiological measurement amplifiers can also be used independently of the acquisition system discussed here. For example, connection to an instrumentation recorder, a polygraph, or other acquisition system can be carried out without problem. The necessary adjustment options are present in the amplifier units, and in general, the signal-output levels are suitable.

The data-acquisition ensemble can be considered as a general-purpose system, suitable for many other applications. For instance, the complete system

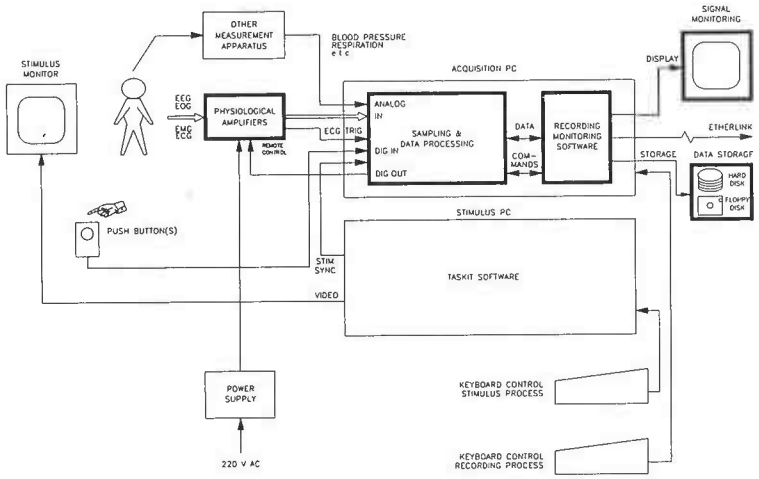


Figure 2. A complete set-up for ERP research.

can be used as a replacement of analog tape-recording units, which not only offers a much better signal-to-noise ratio, but also the additional advantage of recording digital events synchronously at the analog channels.

Depending on the data-reduction requirements, such a system which includes the possibility of optical data storage, may be less expensive than an analog instrumentation recorder. Furthermore, the system has many advantages with respect to transferring data to other systems, user friendliness, flexibility, and reliability.

A final application to be mentioned is the option of off-line data processing by the DSP processor of the data-acquisition board. Because of the availability of powerful commands for signal processing (filtering, FFT etc.) and the possibility of including such commands in a Pascal or C application program, the system can be used as a powerful signal-processing station.

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